

Mathematical Funditions of Reinforcment Learning and Online Decision Making

Homework Assignment 1

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1 Question 1

1.1 a)

The period of a state x is defined by

$$d(x) = \gcd\{n \geq 1 : P^n(x, x) > 0\}.$$

Since x has self-loop

$$P(x, x) > 0,$$

and we have

$$P^1(x, x) > 0.$$

Therefore, 1 belongs to the set

$$\{n \geq 1 : P^n(x, x) > 0\},$$

and hence

$$d(x) = 1.$$

Thus, the state x is aperiodic.

Now let $y \in X$ be any other state. Since the Markov chain is irreducible, there exist integers $m, n \geq 1$ such that

$$P^m(y, x) > 0 \quad \text{and} \quad P^n(x, y) > 0.$$

Because $P(x, x) > 0$, we can remain at x for any additional number of steps. Hence, for every $k \geq 0$, there is a path from y back to y of length

$$m + k + n.$$

Therefore,

$$P^{m+k+n}(y, y) > 0 \quad \text{for every } k \geq 0.$$

In particular,

$$P^{m+n}(y, y) > 0$$

and

$$P^{m+n+1}(y, y) > 0.$$

Consequently,

$$d(y) = \gcd\{r \geq 1 : P^r(y, y) > 0\}$$

divides both $m + n$ and $m + n + 1$. Since two consecutive integers are coprime,

$$\gcd(m + n, m + n + 1) = 1.$$

Thus,

$$d(y) = 1.$$

Since y was arbitrary, every state has period 1. Therefore, the Markov chain is aperiodic.

1.2 b)

Consider a random walk on an odd number of states arranged in a cycle. From each state, the chain moves to either of its two neighboring states with equal probability and there are no self-loops. The chain is irreducible because every state can be reached from every other state. It is also aperiodic because, with an odd number of states, the chain can return to the same state in both an even and an odd number of steps, so the greatest common divisor of the possible return times is 1.

2 Question 2

2.1 a)

Choose a state $x_0 \in X$. Since the chain is irreducible and has period b , partition the state space into b non-overlapping sets $C_1, C_2, \dots, C_b$, where C_i contains the states that can be reached from x_0 in a number of steps congruent to $i - 1$ modulo b . In other words,

$$C_i = \{y \in X : P^s(x_0, y) > 0 \text{ for some } s \equiv i - 1 \pmod{b}\}.$$

Then the transitions must have the form

$$C_1 \rightarrow C_2 \rightarrow \dots \rightarrow C_b \rightarrow C_1.$$

In particular, starting from a state in C_1 , the chain can return to the set C_1 only after $b, 2b, 3b, \dots$ steps. However, returning to the set C_1 does not necessarily mean returning to the same state. Therefore, if we start from a state $x \in C_1$, any return to the same state x must occur after a multiple of b steps, which shows the model is b -periodic.

2.2 b)

The states are ordered according to the sets $C_1, \dots, C_b$, and transitions can only go from C_i to C_{i+1} (with $C_b \rightarrow C_1$). So the transition matrix $P_{n \times n}$ has the following block sparsity structure:

$$P_{n \times n} = \begin{pmatrix} 0 & P_{12} & 0 & \dots & 0 \\ 0 & 0 & P_{23} & \dots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & P_{b-1,b} \\ P_{b1} & 0 & \dots & 0 & 0 \end{pmatrix},$$

where $P_{i,i+1}$ is an $n_i \times n_{i+1}$ block, and P_{b1} is an $n_b \times n_1$ block.

In other words, all other blocks are zero, because there are no transitions between non-consecutive classes.

3 Question 3

3.1 a)

Since P is symmetric, the sum of each column is equal to the sum of each row, which is 1. Therefore, for the uniform distribution $\pi = (1/n, \dots, 1/n) \in \mathbb{R}^n$, we have $\pi P = \pi$. Hence, π is a stationary distribution.

3.2 b)

Yes. If the sum of every column is (1), then the uniform distribution is stationary. Such a matrix can be non-symmetric, for example by reordering the rows of a symmetric transition matrix.

4 Question 4

4.1 a)

Since

$$V_T(x_0) = \frac{1}{T+1} \mathbb{E} \left[\sum_{t=0}^T R(X_t) \mid X_0 = x_0 \right],$$

we have

$$V_T(x_0) = \frac{1}{T+1} \sum_{t=0}^T \sum_{x \in X} P(X_{t+1}, x \mid X_t = x_0) R(x) = \sum_{x \in X} R(x) \frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x).$$

Since the Markov chain is irreducible and aperiodic,

$$P^t(x_0, x) \longrightarrow \pi(x) \quad \text{as } t \rightarrow \infty.$$

By Cesàro convergence, the average also converges to the same limit:

$$\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \longrightarrow \pi(x).$$

Therefore, since X is finite,

$$\begin{aligned} V(x_0) &= \lim_{T \rightarrow \infty} V_T(x_0) \\ &= \sum_{x \in X} \pi(x) R(x). \end{aligned}$$

In particular, the limiting average reward does not depend on the initial state x_0 .

4.2 b)

Let $\tau = \tau_{\text{mix}}(\epsilon)$ and $Y_n = X_{n,\tau}$. Then

$$\widehat{V}_N = \frac{1}{N} \sum_{n=1}^N R(Y_n).$$

We first decompose the MSE. Let $\mu = \mathbb{E}[\widehat{V}_N]$. Then

$$\widehat{V}_N - V = (\widehat{V}_N - \mu) + (\mu - V).$$

Therefore,

$$\begin{aligned} \mathbb{E}[(\widehat{V}_N - V)^2] &= \mathbb{E}[(\widehat{V}_N - \mu)^2] + 2(\mu - V)\mathbb{E}[\widehat{V}_N - \mu] + (\mu - V)^2 \\ &= \mathbb{E}[(\widehat{V}_N - \mu)^2] + (\mathbb{E}[\widehat{V}_N] - V)^2 \\ &= \text{Var}(\widehat{V}_N) + \text{Bias}(\widehat{V}_N)^2. \end{aligned}$$

We now bound the variance and the bias.

Variance. Since the N simulations are independent,

$$\begin{aligned}\text{Var}(\widehat{V}_N) &= \text{Var}\left(\frac{1}{N} \sum_{n=1}^N R(Y_n)\right) \\ &= \frac{1}{N^2} \sum_{n=1}^N \text{Var}(R(Y_n)) \\ &= \frac{1}{N} \text{Var}(R(Y_N)).\end{aligned}$$

By Popoviciu's inequality, if a random variable Z satisfies $m \leq Z \leq M$, then

$$\text{Var}(Z) \leq \frac{1}{4}(M - m)^2.$$

Since

$$-R_{\max} \leq R(Y_N) \leq R_{\max},$$

Therefore,

$$\text{Var}(R(Y_N)) \leq \frac{1}{4}(R_{\max} - (-R_{\max}))^2 = R_{\max}^2.$$

Hence,

$$\text{Var}(\widehat{V}_N) \leq \frac{R_{\max}^2}{N}.$$

Bias. Let $\tau = \tau_{\text{mix}}(\epsilon)$. Since

$$\widehat{V}_N = \frac{1}{N} \sum_{n=1}^N R(Y_n),$$

we have

$$\mathbb{E}[\widehat{V}_N] = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[R(Y_n)].$$

Since all Y_n have the same distribution,

$$\mathbb{E}[\widehat{V}_N] = \mathbb{E}[R(Y_n)].$$

Let $P^\tau(x_0, x) = \mathbb{P}(Y_n = x)$. Then

$$\mathbb{E}[R(Y_n)] = \sum_{x \in X} P^\tau(x_0, x) R(x).$$

From part (a),

$$V = \sum_{x \in X} \pi(x) R(x).$$

Therefore,

$$\begin{aligned}|\text{Bias}(\widehat{V}_N)| &= |\mathbb{E}[\widehat{V}_N] - V| \\ &= \left| \sum_{x \in X} (P^\tau(x_0, x) - \pi(x)) R(x) \right|.\end{aligned}$$

Since

$$\|P^\tau(x_0, \cdot) - \pi\|_{\text{TV}} = \frac{1}{2} \sum_{x \in X} |P^\tau(x_0, x) - \pi(x)| \leq \epsilon$$

and $|R(x)| \leq R_{\max}$, we obtain

$$|\text{Bias}(\widehat{V}_N)| \leq 2R_{\max}\epsilon.$$

Thus,

$$\text{Bias}(\widehat{V}_N)^2 \leq 4R_{\max}^2 \epsilon^2.$$

Finally,

$$\begin{aligned} \mathbb{E}[(\widehat{V}_N - V)^2] &\leq \frac{R_{\max}^2}{N} + 4R_{\max}^2 \epsilon^2 \\ &= R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right). \end{aligned}$$

4.3 c)

From part (b), we have the bound

$$\mathbb{E}[(\widehat{V}_N - V)^2] \leq R_{\max}^2 \left(\frac{1}{N} + 4\delta^2 \right),$$

where δ is the accuracy used in the mixing time, i.e.,

$$\|P^\tau(x_0, \cdot) - \pi\|_{\text{TV}} \leq \delta.$$

In part (c), however, ϵ represents the **final MSE accuracy** that we want:

$$\mathbb{E}[(\widehat{V}_N - V)^2] \leq \epsilon.$$

We now choose N and δ so that both variance and bias terms in the bound from part (b) each contribute at most $\epsilon/2$.

First, choose N such that

$$\frac{R_{\max}^2}{N} \leq \frac{\epsilon}{2}.$$

This gives

$$N \geq \frac{2R_{\max}^2}{\epsilon}.$$

Next, choose δ such that

$$4R_{\max}^2 \delta^2 \leq \frac{\epsilon}{2}.$$

Solving for δ ,

$$\delta^2 \leq \frac{\epsilon}{8R_{\max}^2},$$

so we choose

$$\delta \leq \sqrt{\frac{\epsilon}{8R_{\max}^2}}.$$

With these choices,

$$\begin{aligned} \mathbb{E}[(\widehat{V}_N - V)^2] &\leq R_{\max}^2 \left(\frac{1}{2R_{\max}^2/\epsilon} + 4\frac{\epsilon}{8R_{\max}^2} \right) \\ &= R_{\max}^2 \left(\frac{\epsilon}{2R_{\max}^2} + \frac{\epsilon}{2R_{\max}^2} \right) \\ &= \epsilon. \end{aligned}$$

Therefore, these choices guarantee that $\widehat{V}_N$ is ϵ -accurate with respect to the MSE. It remains to determine how many steps of the Markov chain are needed for each simulation. Since

$$\delta^{-1} = \sqrt{\frac{8R_{\max}^2}{\epsilon}}.$$

Using the hint,

$$\tau_{\text{mix}}(\delta) \leq \lceil \log_2(\delta^{-1}) \rceil \tau_{\text{mix}}(1/4).$$

Substituting the value of δ^{-1} ,

$$\begin{aligned} \tau_{\text{mix}}(\delta) &\leq \left\lceil \log_2 \left(\sqrt{\frac{8R_{\max}^2}{\epsilon}} \right) \right\rceil \tau_{\text{mix}}(1/4) \\ &= \left\lceil \frac{1}{2} \log_2 \left(\frac{8R_{\max}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}}(1/4). \end{aligned}$$

Therefore, each of the N independent simulations needs at most

$$\left\lceil \frac{1}{2} \log_2 \left(\frac{8R_{\max}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}}(1/4)$$

steps.

Since

$$N \geq \frac{2R_{\max}^2}{\epsilon},$$

the total number of Markov chain steps, $N \times \tau_{\text{mix}}(\delta)$, is

$$\left\lceil \frac{2R_{\max}^2}{\epsilon} \right\rceil \left\lceil \frac{1}{2} \log_2 \left(\frac{8R_{\max}^2}{\epsilon} \right) \right\rceil \tau_{\text{mix}}(1/4).$$

I think there is a typo in the assignment draft, as it is not multiplied by 2.

5 Question 5

5.1 a)

Suppose x_1^* and x_2^* are two fixed points of T . Then

$$T(x_1^*) = x_1^*, \quad T(x_2^*) = x_2^*.$$

Since T is an α -contraction,

$$\begin{aligned} \|x_1^* - x_2^*\| &= \|T(x_1^*) - T(x_2^*)\| \\ &\leq \alpha \|x_1^* - x_2^*\|. \end{aligned}$$

Since $0 \leq \alpha < 1$, this is only possible if

$$\|x_1^* - x_2^*\| = 0.$$

Therefore,

$$\boxed{x_1^* = x_2^*}.$$

Hence, the fixed point is unique.

5.2 b)

Let

$$T(x) = \frac{x}{2} + 1, \quad X = (0, 1)$$

Then, for any $x, y \in (0, 1)$,

$$|T(x) - T(y)| = \left| \frac{x}{2} + 1 - \left(\frac{y}{2} + 1 \right) \right| = \frac{1}{2}|x - y|.$$

Thus, T is an α -contraction with

$$\frac{1}{2} \leq \alpha < 1.$$

However, T has no fixed point in the domain. Indeed, if x^* were a fixed point, then

$$T(x^*) = x^*,$$

so

$$\frac{x^*}{2} + 1 = x^*.$$

This gives

$$x^* = 2.$$

That is out of the domain.

5.3 c)

Let $x_0 \in X$ be arbitrary and define the sequence

$$x_{n+1} = T(x_n), \quad n \geq 0.$$

We will show that (x_n) is a Cauchy sequence. Since X is complete, it will then have a limit $x^* \in X$.

Because T is an α -contraction,

$$\|x_{n+1} - x_n\| = \|T(x_n) - T(x_{n-1})\| \leq \alpha^n \|x_1 - x_0\|.$$

Now, for $m > n$, by the triangle inequality,

$$\begin{aligned} \|x_m - x_n\| &\leq \|x_m - x_{m-1}\| + \cdots + \|x_{n+1} - x_n\| \\ &\leq \sum_{k=n}^{m-1} \alpha^k \|x_1 - x_0\| \\ &= \|x_1 - x_0\| \sum_{k=n}^{m-1} \alpha^k. \end{aligned}$$

Since $0 < \alpha < 1$,

$$\sum_{k=n}^{m-1} \alpha^k \leq \sum_{k=n}^{\infty} \alpha^k = \frac{\alpha^n}{1 - \alpha}.$$

Therefore,

$$\|x_m - x_n\| \leq \frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\|.$$

As $n \rightarrow \infty$, since $0 < \alpha < 1$, for all $n < m$

$$\frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\| \rightarrow 0 \Rightarrow \|x_m - x_n\| \rightarrow 0 \Rightarrow x_m = x_n.$$

Hence (x_n) is a Cauchy sequence. Since X is complete,

$$x_n = x_{n+1} = T(x_n) \Rightarrow x_n = x^*; \quad \text{as } n \rightarrow \infty.$$

Finally,

$$T(x^*) = x^*.$$

Therefore, T has a fixed point in X .

6 Question 6

6.1 a)

Let $T(x) = Wx$. Suppose, for contradiction, that T is an α -contraction in some norm, where $0 \leq \alpha < 1$.

Since W is doubly stochastic, we have

$$W\mathbf{1} = \mathbf{1}.$$

Take $x = \mathbf{1}$ and $y = 0$. Then

$$T(\mathbf{1}) = W\mathbf{1} = \mathbf{1}, \quad T(0) = W0 = 0.$$

Therefore,

$$\begin{aligned} \|T(\mathbf{1}) - T(0)\| &= \|\mathbf{1} - 0\| \\ &= \|\mathbf{1}\|. \end{aligned}$$

Since T is assumed to be an α -contraction,

$$\|T(\mathbf{1}) - T(0)\| \leq \alpha \|\mathbf{1} - 0\|.$$

Hence,

$$\|\mathbf{1}\| \leq \alpha \|\mathbf{1}\|.$$

this gives $1 \leq \alpha$, which contradicts $\alpha < 1$. Therefore, $x \mapsto Wx$ is not contractive in any norm.

6.2 b)

Suppose x is a fixed point. Then

$$Wx = x.$$

Thus, x is an eigenvector of W corresponding to the eigenvalue 1.

Since W is an irreducible non-negative matrix and its spectral radius is

$$\rho(W) = 1,$$

the Perron–Frobenius theorem tells us that the eigenspace corresponding to the eigenvalue 1 is one-dimensional.

Also, since W is doubly stochastic,

$$W\mathbf{1} = \mathbf{1}.$$

Therefore, $\mathbf{1}$ is an eigenvector corresponding to the eigenvalue 1.

Since the eigenspace is one-dimensional, the only eigenvectors corresponding to 1 are multiples of $\mathbf{1}$. Hence,

$$x = a\mathbf{1}$$

for some $a \in \mathbb{R}$.

Therefore, the set of fixed points is

$$\text{Span}(\mathbf{1}) = \{a\mathbf{1} \in \mathbb{R}^N : a \in \mathbb{R}\}.$$

6.3 c)

Let

$$A(\bar{x}) = \left\{ x \in \mathbb{R}^N : \frac{1}{N}\mathbf{1}^T x = \bar{x} \right\}.$$

Suppose $x \in A(\bar{x})$. Then

$$\frac{1}{N}\mathbf{1}^T x = \bar{x}.$$

Since W is doubly stochastic,

$$\mathbf{1}^T W = \mathbf{1}^T.$$

Therefore,

$$\begin{aligned} \frac{1}{N}\mathbf{1}^T(Wx) &= \frac{1}{N}(\mathbf{1}^T W)x \\ &= \frac{1}{N}\mathbf{1}^T x \\ &= \bar{x}. \end{aligned}$$

Hence,

$$Wx \in A(\bar{x}).$$

6.4 d)

Let

$$A(\bar{x}) = \left\{ x \in \mathbb{R}^N : \frac{1}{N}\mathbf{1}^T x = \bar{x} \right\}.$$

Suppose $x \in A(\bar{x})$. Then

$$\mathbf{1}^T x = N\bar{x}.$$

Consider

$$x - \bar{x}\mathbf{1}.$$

This represents the difference between the values at the nodes and their average. We have

$$\begin{aligned}\mathbf{1}^T(x - \bar{x}\mathbf{1}) &= \mathbf{1}^T x - \bar{x}\mathbf{1}^T \mathbf{1} \\ &= N\bar{x} - N\bar{x} \\ &= 0.\end{aligned}$$

Thus, $x - \bar{x}\mathbf{1}$ is orthogonal to $\mathbf{1}$.

Since $W\mathbf{1} = \mathbf{1}$,

$$W(\bar{x}\mathbf{1}) = \bar{x}\mathbf{1}.$$

Therefore,

$$\begin{aligned}Wx - \bar{x}\mathbf{1} &= Wx - W(\bar{x}\mathbf{1}) \\ &= W(x - \bar{x}\mathbf{1}).\end{aligned}$$

Now define

$$J = \frac{1}{N}\mathbf{1}\mathbf{1}^T.$$

Since $x - \bar{x}\mathbf{1}$ has average zero,

$$J(x - \bar{x}\mathbf{1}) = 0.$$

Hence,

$$W(x - \bar{x}\mathbf{1}) = (W - J)(x - \bar{x}\mathbf{1}).$$

Therefore,

$$\begin{aligned}\|Wx - \bar{x}\mathbf{1}\|_2 &= \|(W - J)(x - \bar{x}\mathbf{1})\|_2 \\ &\leq \|W - J\|_2 \|x - \bar{x}\mathbf{1}\|_2.\end{aligned}$$

Define

$$\zeta = \left\| W - \frac{1}{N}\mathbf{1}\mathbf{1}^T \right\|_2.$$

Because W is irreducible and non-negative, $W\mathbf{1} = \mathbf{1}$, and $W_{nn} > 0$, the Perron–Frobenius theorem implies that 1 is the unique eigenvalue on the unit circle and all other eigenvalues have magnitude strictly less than 1. Thus, on the subspace orthogonal to $\mathbf{1}$, W is strictly contractive, giving

$$\zeta < 1.$$

Hence,

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \zeta \|x - \bar{x}\mathbf{1}\|_2, \quad \zeta = \left\| W - \frac{1}{N}\mathbf{1}\mathbf{1}^T \right\|_2 < 1.$$

Therefore, W reduces the distance to the average vector $\bar{x}\mathbf{1}$ by a factor smaller than 1. Thus, W is a pseudo-contraction on $A(\bar{x})$.

6.5 e)

Since W is symmetric, it has an orthonormal basis of eigenvectors

$$v_1, v_2, \dots, v_N.$$

Because

$$W\mathbf{1} = \mathbf{1},$$

we can take

$$v_1 = \frac{1}{\sqrt{N}}\mathbf{1}$$

with eigenvalue

$$\lambda_1 = 1.$$

Write the initial vector $x(0)$ in this eigenvector basis:

$$x(0) = \sum_{i=1}^N a_i v_i.$$

Since

$$x(k) = W^k x(0),$$

we have

$$x(k) = \sum_{i=1}^N a_i W^k v_i.$$

Because v_i is an eigenvector with eigenvalue λ_i ,

$$W^k v_i = \lambda_i^k v_i.$$

Therefore,

$$x(k) = \sum_{i=1}^N a_i \lambda_i^k v_i.$$

The first term is

$$a_1 v_1 = \frac{1}{N} \mathbf{1}^T x(0) \mathbf{1} = \bar{x} \mathbf{1}.$$

Hence,

$$x(k) - \bar{x} \mathbf{1} = \sum_{i=2}^N a_i \lambda_i^k v_i.$$

Since the eigenvectors are orthonormal,

$$\|x(k) - \bar{x} \mathbf{1}\|_2^2 = \sum_{i=2}^N a_i^2 |\lambda_i|^{2k}.$$

By definition, λ_2 is the second largest eigenvalue in absolute value, so

$$|\lambda_i| \leq |\lambda_2|, \quad i \geq 2.$$

Thus,

$$\|x(k) - \bar{x}\mathbf{1}\|_2^2 \leq |\lambda_2|^{2k} \sum_{i=2}^N a_i^2.$$

Taking square roots,

$$\|x(k) - \bar{x}\mathbf{1}\|_2 \leq \sqrt{\sum_{i=2}^N a_i^2} |\lambda_2|^k.$$

Define

$$A = \sqrt{\sum_{i=2}^N a_i^2}.$$

Therefore,

$$\|x(k) - \bar{x}\mathbf{1}\|_2 \leq A |\lambda_2|^k.$$

By the Perron–Frobenius theorem,

$$|\lambda_2| < 1.$$

Hence,

$$\|x(k) - \bar{x}\mathbf{1}\|_2 \rightarrow 0$$

at a linear (geometric) rate determined by the second largest eigenvalue in absolute value.

7 Question 7

Consider the gradient descent algorithm

$$x(k+1) = x(k) - \gamma (\nabla f(x(k)) + e(k)),$$

where $e(k)$ is IID zero-mean noise with

$$\mathbb{E}[e(k)] = 0, \quad \mathbb{E}\|e(k)\|^2 = \sigma^2.$$

Assume that f is L -smooth and μ -convex, and let $x^\star$ be the minimizer of f . We assume

$$0 < \gamma < \frac{2}{L + \mu}.$$

We want to show that

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)^{k+1} \mathbb{E}\|x(0) - x^\star\|^2 + \frac{\gamma\sigma^2(\mu + L)}{2\mu L}.$$

First, we write

$$x(k+1) - x^\star = x(k) - x^\star - \gamma (\nabla f(x(k)) + e(k)).$$

Therefore,

$$\begin{aligned}
\|x(k+1) - x^\star\|^2 &= \|x(k) - x^\star - \gamma \nabla f(x(k)) - \gamma e(k)\|^2 \\
&= \|x(k) - x^\star - \gamma \nabla f(x(k))\|^2 \\
&\quad - 2\gamma \langle x(k) - x^\star - \gamma \nabla f(x(k)), e(k) \rangle \\
&\quad + \gamma^2 \|e(k)\|^2.
\end{aligned}$$

Taking expectations, the cross term disappears because $e(k)$ is zero mean and independent of $x(k)$. Hence,

$$\mathbb{E}\|x(k+1) - x^\star\|^2 = \mathbb{E}\|x(k) - x^\star - \gamma \nabla f(x(k))\|^2 + \gamma^2 \sigma^2.$$

Expanding the first term gives

$$\begin{aligned}
\mathbb{E}\|x(k+1) - x^\star\|^2 &= \mathbb{E}\|x(k) - x^\star\|^2 \\
&\quad - 2\gamma \mathbb{E}[\langle \nabla f(x(k)), x(k) - x^\star \rangle] \\
&\quad + \gamma^2 \mathbb{E}\|\nabla f(x(k))\|^2 + \gamma^2 \sigma^2.
\end{aligned}$$

From the given inequality, taking $y = x^\star$ and using

$$\nabla f(x^\star) = 0,$$

we have

$$\langle \nabla f(x(k)), x(k) - x^\star \rangle \geq \frac{\mu L}{\mu + L} \|x(k) - x^\star\|^2 + \frac{1}{\mu + L} \|\nabla f(x(k))\|^2.$$

Substituting this inequality gives

$$\begin{aligned}
\mathbb{E}\|x(k+1) - x^\star\|^2 &\leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right) \mathbb{E}\|x(k) - x^\star\|^2 \\
&\quad + \left(\gamma^2 - \frac{2\gamma}{\mu + L}\right) \mathbb{E}\|\nabla f(x(k))\|^2 + \gamma^2 \sigma^2.
\end{aligned}$$

Since

$$0 < \gamma < \frac{2}{\mu + L},$$

we have

$$\gamma^2 - \frac{2\gamma}{\mu + L} \leq 0.$$

Therefore, we can drop the non-positive term and obtain

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right) \mathbb{E}\|x(k) - x^\star\|^2 + \gamma^2 \sigma^2.$$

Define

$$q = 1 - \frac{2\gamma\mu L}{\mu + L} < 1.$$

Then

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq q \mathbb{E}\|x(k) - x^\star\|^2 + \gamma^2 \sigma^2.$$

Applying this recursion repeatedly,

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq q^{k+1} \mathbb{E}\|x(0) - x^\star\|^2 + \gamma^2 \sigma^2 \sum_{i=0}^k q^i.$$

Using

$$\sum_{i=0}^k q^i = \frac{1 - q^{k+1}}{1 - q} \leq \frac{1}{1 - q},$$

we obtain

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq q^{k+1} \mathbb{E}\|x(0) - x^\star\|^2 + \frac{\gamma^2 \sigma^2}{1 - q}.$$

Since

$$1 - q = \frac{2\gamma\mu L}{\mu + L},$$

we get

$$\frac{\gamma^2 \sigma^2}{1 - q} = \frac{\gamma \sigma^2 (\mu + L)}{2\mu L}.$$

Therefore,

$$\mathbb{E}\|x(k+1) - x^\star\|^2 \leq \left(1 - \frac{2\gamma\mu L}{\mu + L}\right)^{k+1} \mathbb{E}\|x(0) - x^\star\|^2 + \frac{\gamma \sigma^2 (\mu + L)}{2\mu L}.$$

Since

$$0 < 1 - \frac{2\gamma\mu L}{\mu + L} < 1,$$

the first term converges to zero as $k \rightarrow \infty$. Thus,

$$\limsup_{k \rightarrow \infty} \mathbb{E}\|x(k) - x^\star\|^2 \leq \frac{\gamma \sigma^2 (\mu + L)}{2\mu L}.$$

The remaining error is proportional to γ . Hence, by choosing γ sufficiently small, the solution accuracy can be made arbitrarily good, even when $\sigma^2 > 0$.

However, there is a trade-off. A smaller γ makes the error floor smaller, but also makes

$$1 - \frac{2\gamma\mu L}{\mu + L}$$

closer to 1. Therefore, convergence becomes slower.

Hence, the cost of obtaining higher solution accuracy is that more iterations are required.

8 Question 8

An example is the following.

Let

$$\omega \sim \text{Uniform}[0, 1].$$

Divide the interval $[0, 1]$ into smaller and smaller intervals. For example, first use

$$[0, 1],$$

then

$$\left[0, \frac{1}{2}\right), \quad \left[\frac{1}{2}, 1\right),$$

then divide $[0, 1]$ into 4 intervals, then 8 intervals, and so on.

Define $X_k(\omega) = 1$ if ω belongs to the k -th interval, and $X_k(\omega) = 0$ otherwise.

The length of the interval containing $X_k = 1$ becomes smaller and smaller as k increases. Therefore,

$$\mathbb{P}(X_k = 1) \rightarrow 0.$$

For any $0 < \varepsilon < 1$,

$$\mathbb{P}(|X_k - 0| \geq \varepsilon) = \mathbb{P}(X_k = 1) \rightarrow 0.$$

However, for every $\omega \in [0, 1]$, the point ω belongs to one interval at every level. Therefore, $X_k(\omega) = 1$ infinitely many times. It also equals 0 infinitely many times.

Hence $X_k(\omega)$ does not converge to 0. Therefore,

$$X_k \not\xrightarrow{\text{a.s.}} 0.$$

Thus, we have an example of a sequence that converges in probability but does not converge almost surely:

$$X_k \xrightarrow{P} 0 \quad \text{but} \quad X_k \not\xrightarrow{\text{a.s.}} 0.$$

9 Question 9

9.1 a)

By Markov's inequality,

$$P(|X - \mu| \geq \epsilon) = P((X - \mu)^2 \geq \epsilon^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\epsilon^2}.$$

Since

$$\mathbb{E}[(X - \mu)^2] = \text{Var}(X),$$

we get

$$P(|X - \mu| \geq \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}.$$

9.2 b)

Since the X_i are IID,

$$\mathbb{E}[S_n] = \mu, \quad \text{Var}(S_n) = \frac{\sigma^2}{n}.$$

By Chebyshev's inequality,

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(S_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}.$$

Since

$$\frac{\sigma^2}{n\epsilon^2} \rightarrow 0,$$

we have

$$P(|S_n - \mu| \geq \epsilon) \rightarrow 0.$$

Therefore,

$$S_n \xrightarrow{P} \mu.$$

9.3 c)

Using Chebyshev's inequality,

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

We want

$$P(|S_n - \mu| \leq \epsilon) \geq 1 - \delta.$$

So choose n such that

$$\frac{\sigma^2}{n\epsilon^2} \leq \delta.$$

Therefore,

$$n \geq \frac{\sigma^2}{\delta\epsilon^2}.$$

Thus, you need at least

$$\left\lceil \frac{\sigma^2}{\delta\epsilon^2} \right\rceil$$

samples.